

A Hybrid Islanding Detection Technique for Inverter Based Distributed Generations

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Abstract—This paper proposes a new hybrid islanding detection method for grid-connected inverter based distributed generation units. The proposed method combines passive and active islanding detection techniques to aggregate their individual advantages and reduce or eliminate their drawbacks. The passive method is based on extraction of critical system parameters from local voltage measurements at target DG location and application of decision tree based classifiers for extraction of decision rules in order to detect islanding events. The active method is based on Sandia Frequency Shift technique and is initiated only when the passive method is unable to properly classify between islanding and other transient generating events, thus minimizing the power quality degradation introduced in the system. A detailed case study on a grid-connected PV system is performed to evaluate the performance of the proposed technique. Test results demonstrate the effectiveness of the proposed method in detection of islanding events in grid-connected distributed generations.

Index Terms—Distributed generations, decision trees, Sandia frequency shift, islanding detection, electric power systems.

I. INTRODUCTION

Integration of distributed generations (DGs) with area electric power systems (EPS) offers numerous benefits including potential solution to energy security and reliability with minimum environmental impacts. However, several technical considerations are required in system planning and operation processes for grid-integration of DGs. Inadvertent islanding is one of the major issues associated with integration of DGs with area EPS. IEEE Std. 1547 defines islanding as a condition in which a portion of an area electric power system (EPS) is energized solely by one or more local EPSs through the associated points of common couplings (PCCs) while that portion of the area EPS is electrically separated from the rest of area EPS [1]. IEEE Std. 1547 recommends isolation of DGs within a maximum of 2 seconds in events of island formation.

Although islanded operations may be able to enhance reliability by supplying local loads and reducing downtime when supply from area EPS is unavailable, but several operational and safety considerations including power quality standards, voltage and frequency controls, and safety hazards are required before such operations can be realized in practice. Distribution systems are traditionally designed as radial systems and time coordination between protective devices is the standard practice among utilities for secured and reliable system operation. Integration of DG resources downstream of protective devices requires fast and accurate islanding detection as the primary step for secured operation [2-3].

Islanding detection techniques are classified into remote and local techniques. Remote techniques are communication based techniques depending on power line signaling and transfer trip methods. Although these techniques are reliable but their expensive nature makes them uneconomical from DG standpoint. Local techniques utilize measurements at target DG location for islanding detection. Local techniques can be sub-divided into passive and active techniques.

Active techniques rely upon perturb and observe methods. In grid-connected mode, injected perturbations produce insignificant variations in system parameters, whereas significantly larger variations are observed if the DG is operating in islanded mode. Several active islanding detection techniques have been proposed in literature [4-6] including active frequency drift [4], slip-mode frequency shift [5], Sandia frequency shift [6] and Sandia voltage shift [6]. Active methods have smaller non-detection zone but degradation in power quality caused by perturbation injections is a major drawback.

Passive islanding detection methods rely on local measurements of critical system parameters (such as voltage, current, and frequency etc.) and detects islanding events by locating abnormalities in those system parameters. Passive islanding detection methods proposed in literature include over/under voltage [7], over/under frequency [7], rate-of-change of frequency [8], rate-of-change of voltage [9], voltage unbalance and total harmonic distortion [10] based methods etc. Several methods based on data mining approaches have also been proposed in literature [11-13]. Passive islanding detection methods do not degrade power quality, but these methods suffer from larger non-detection zone (NDZ). Especially, in presence of power balance in the island (i.e. generation and load are approximately balanced in the islanded system).

Hybrid islanding detection methods based on voltage unbalance/ frequency shift and average rate-of-change of voltage/real power shift are also proposed in literature [14-15]. Both of the methods considers only synchronous/asynchronous DGs (i.e. diesel generators, wind turbine generators etc.) and requires manual selection of the threshold values based on designer/operator experience for the passive detection scheme.

This paper proposes a hybrid islanding detection method for inverter based DGs. The proposed method combines the principles of decision trees (DTs) for passive islanding detection and Sandia frequency shift (SFS) for active islanding detection in order to enhance detection accuracy and reliability by reducing NDZ and degradation in power quality. Case

study on a 200 kW grid-connected photovoltaic (PV) farm is performed to verify the effectiveness of the proposed method.

The paper is organized as follows. Principles of the proposed hybrid islanding detection method is presented in Section II. Section III describes the test system model. In Section IV, case study results are presented, and performance of the proposed hybrid islanding detection method is discussed. Finally, Conclusions are presented in Section V.

II. PROPOSED METHODOLOGY

The proposed hybrid islanding detection method is composed of two main components: (1) passive islanding based on decision trees (DT) and (2) active islanding detection based on Sandia frequency shift (SFS) method. An overview of the proposed methodology is presented in this section.

A. Decision Tree Based Passive IDM

The passive islanding detection method is based on extraction of system parameters by post-processing voltage waveforms obtained from measurements at target DG locations, and detection of event specific signatures on these system parameters through application of DT based pattern recognition technique. Table I lists the system parameters (features) used for passive islanding detection.

TABLE I. PARAMETERS USED IN PASSIVE ISLANDING DETECTION

$x_1^i = \Delta V_i$	Voltage deviation under i^{th} event (Volts)
$x_2^i = \left(\frac{\Delta V}{\Delta t}\right)_i$	Rate-of-change of voltage under i^{th} event (Volts/cycle)
$x_3^i = \left(\frac{V_2}{V_1}\right)_i$	Voltage unbalance under i^{th} event
$x_4^i = V_{THD i}$	Total harmonic distortion in voltage under i^{th} event

The features are selected considering the facts that inverter based DGs for utility-interconnected services generally operate close to unity power factor. Hence islanding events may lead to possible deficiency in reactive power and subsequent voltage variations which can be utilized for islanding detection. Moreover, islanding scenarios may also lead to variations in network topology and DG loading conditions which might lead to variations in voltage unbalance and total harmonic distortions in voltage waveforms generated by the inverter.

In the proposed scheme, the DT based passive method consists of four major components- (1) Offline DT training, (2) DT based threshold setting, (3) periodic DT update and (4) online application.

1) *Offline Decision Tree training*: The DTs built in this proposed scheme are classification trees built from training dataset consisting of feature vectors and corresponding class values. Properly trained DTs can predict the class of an event (i.e. non-islanding or islanding events) from the feature vector presented to it for classification. The feature vector consists of system parameter values utilized in the classification process. The proposed approach can be mathematically illustrated as:

$$X_i = \{x_1^i, x_2^i, x_3^i, x_4^i\}; i = 1, 2, 3, \dots, N \quad (1)$$

$$F = [X_1, X_2, \dots, X_N]^T \quad (2)$$

$$Y = [y^1, y^2, \dots, y^N]^T \quad (3)$$

where X_i is the feature vector containing feature values for i^{th} event, F is the matrix containing feature vectors for each

of the N system events, Y contains the class values (i.e. $y_i = 0$ for non-islanding events and $y_i = 1$ for islanding events) corresponding to the feature vectors X_i for each of the N system events. The complete data model (E) for DT classifier training can be expressed as follows:

$$E = [F \ Y] = \begin{bmatrix} x_1^1 & x_2^1 & x_3^1 & x_4^1 & y^1 \\ x_1^2 & x_2^2 & x_3^2 & x_4^2 & y^2 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ x_1^N & x_2^N & x_3^N & x_4^N & y^N \end{bmatrix} \quad (4)$$

Training dataset containing feature vectors and corresponding class values are obtained from extensive offline simulation of prescribed system events. A wide variety of system disturbances leading to islanding and non-islanding events are considered in offline simulations for acquiring essential system characteristics. Moreover, these events are simulated under various operating states of area EPS and DG unit.

2) *Decision Tree Based Threshold Setting*: The DT based islanding classification model yields a set of decision rules associated with the classification process. Decision rules are essentially decision boundaries in the data model which differentiates islanding events from non-islanding events. Threshold values for system parameters associated with the classification process are obtained from the decision rules generated from offline DT building process.

In the proposed scheme, two sets of threshold values were generated- (1) Lower threshold setting (TH_L): differentiates normal operating conditions from other system events, and (2) upper threshold settings (TH_U): differentiates islanding events from other system events. An event is classified as non-islanding event only if the selected feature values are below the lower threshold settings, TH_L . Whereas, an event is classified as an islanding event only when the feature values are above the upper threshold settings, TH_U . If the feature values are between TH_L and TH_U , the active islanding detection method based on Sandia frequency shift (SFS) is initiated.

3) *Periodic DT Update*: The DTs developed offline can be updated periodically by incorporating the changing system states in the DT training process. If significant changes (in load levels, generation status etc.) exist, then the performance of the existing DTs are evaluated under the new operating conditions. If the DTs do not perform satisfactorily, then offline event simulations are carried out under the new operating conditions. DTs are trained with the enhanced training set containing the new cases alongside the original cases and updated threshold settings are obtained.

4) *Online Implementation*: Voltage measurements at target DG location are obtained in continuously in real time. Feature vector containing the critical system parameter values are calculated, compared with the threshold settings and necessary security measure is appropriated.

B. Active IDM Based on Sandia Frequency Shift

Sandia frequency shift (SFS) method [6] is essentially an active frequency drifting (AFD) [4] islanding detection method with positive feedback. SFS method injects small perturbations in inverter output current and applies positive feedback to destabilize system frequency. In grid-connected mode, the injected perturbations produces insignificant variations in system

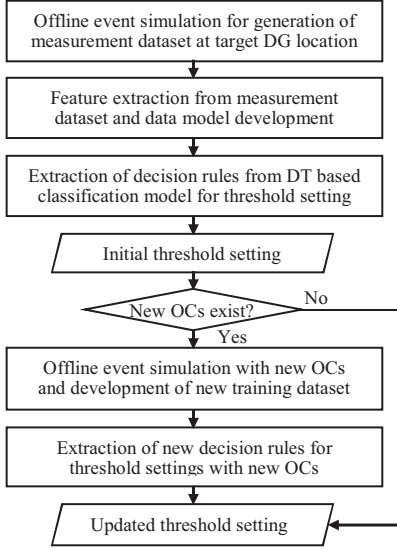


Figure 1. Flowchart of the proposed passive islanding detection scheme.

frequency. Whereas, in islanded mode, significantly larger frequency variations are caused by injected perturbations. Perturbations are injected in inverter output current as one zero time (t_z) segment per line semi-cycle. The ratio of the zero time (t_z) to half of the period of voltage waveform (T_v) as expressed in (5), is called the chopping fraction (cf). Positive feedback formulated as a linear function of deviation in frequency of PCC voltage is applied to chopping fraction (cf) which causes the frequency to deviate from the nominal value in the absence of utility grid. Mathematically, the computation of chopping fraction with positive feedback can be expressed as (6).

$$cf = \frac{2t_z}{T_v} = cf_0 + k(f - f_0) \quad (5)$$

where cf_0 is the chopping fraction when there is no frequency error, k is the accelerating gain, f is the system frequency and f_0 is the nominal value of system frequency in grid-connected mode. Insertion of the zero time (t_z) segment in inverter current advances the phase of fundamental component of inverter current with respect to voltage at PCC giving rise to a phase error and thereby causing frequency to drift away from the nominal value in absence of the grid. Inverter phase angle (θ_{inv}) for SFS method can be computed as:

$$\theta_{inv} = \frac{\omega t_z}{2} = \frac{\pi cf}{2} = \frac{\pi}{2}(cf_0 + k(f - f_0)) \quad (6)$$

SFS can be implemented using phase angle transformation in inverter control block [16]. In general, SFS method is implemented in conjunction with OFP/UFPP relays. Outline of the proposed methodology is presented in Fig. 2.

III. TEST SYSTEM MODEL

The test system model, presented in Fig. 3, consists of a 200 kW inverter based DG unit (PV farm) and loads connected to an equivalent 120 kV transmission network via a 25 kV distribution feeder. The PV farm consists of two PV modules each delivering a maximum power of 100 kW at an irradiance

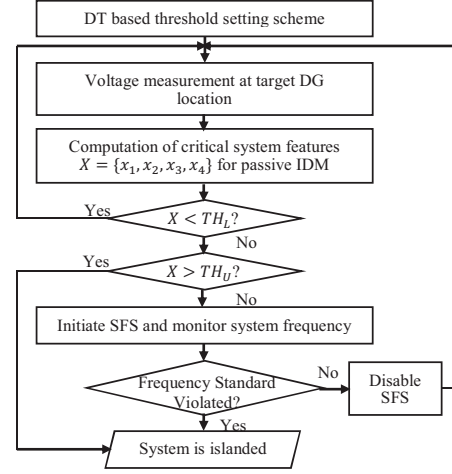


Figure 2. Outline of the proposed hybrid islanding detection method.

of 1000 W/m². The DG unit is connected to the utility grid through a 200 kVA, 260V/25 kV coupling transformer. Loads in the test system model are represented as constant RLC loads.

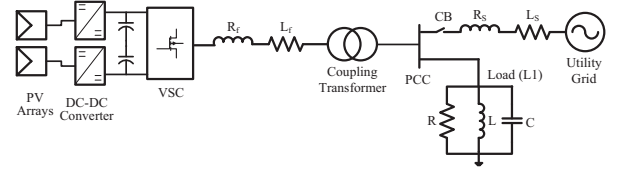


Figure 3. Single line diagram of the test system model.

Fig. 4 presents the implementation of the DT based passive IDM which generates an initiation signal for SFS scheme when parameter values satisfy $TH_L < x_i < TH_U$. Otherwise, if the parameter values are $x_i \geq TH_U$, a trip signal is generated in order to isolate the DG from the distribution system.

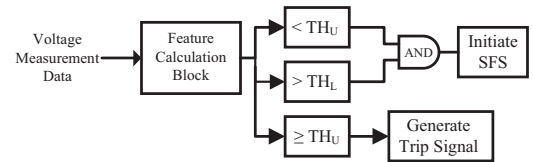


Figure 4. Implementation of DT based passive scheme.

Fig. 5 presents the SFS implementation using phase angle transformation method to obtain new reference values for d-q axis currents i_{dref}^* and i_{qref}^* respectively.

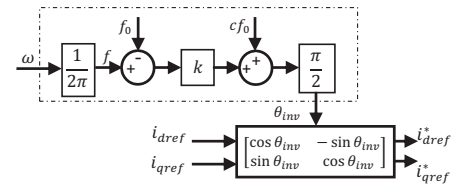


Figure 5. Implementation of SFS scheme in DG interface control.

Implementation of the proposed hybrid IDM in DG interface control architecture is presented in Fig. 6.

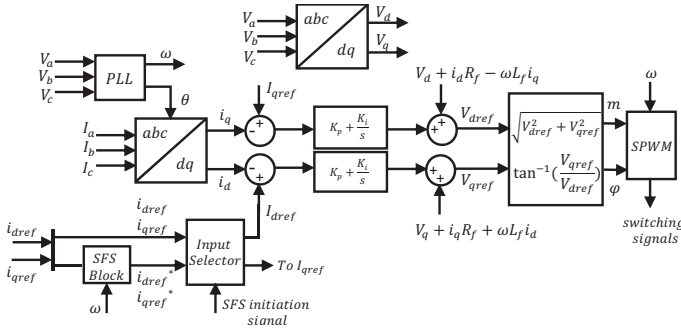


Figure 6. Implementation of the proposed hybrid islanding detection method in DG interface control architecture.

IV. CASE STUDY RESULTS AND ANALYSIS

In this case study, a total of 150 system events under various system operating conditions have been used to generate the measurement database at the target DG location (at the point of common coupling of the DG unit with utility grid). Critical system features are extracted from obtained measurement dataset and data model for setting the threshold values for passive islanding relay at the target DG location is developed. Based on the developed data model, decision trees are constructed using the J48 decision tree algorithm in Waikato Environment for Knowledge Analysis (WEKA) [17] data mining platform. The decision rules and corresponding threshold values are extracted for implementation of the passive islanding relay in the proposed hybrid scheme.

Two sets of decision trees were constructed for extracting the lower and upper threshold settings. The lower threshold settings were obtained by training the decision tree classifiers using all 150 cases generated through system event simulations under both low real power (0 to 6 percent) and reactive power (-1 to 1 percent) mismatch scenarios as well as large real power (greater than 25 percent) and reactive power (-5 to 5 percent) mismatch scenarios between the DG unit and local loads. Decision trees were trained with 85 training cases corresponding to large real and reactive power mismatch scenarios for obtaining the upper threshold settings. Table II presents the threshold settings for the passive islanding relay in the proposed hybrid method for the system under study.

TABLE II. THRESHOLD SETTING FOR THE PASSIVE IDM

Feature	Threshold Values(TH_L)	Threshold Values(TH_U)
Voltage deviation (ΔV)	58.468	243.40
Rate-of-change of voltage ($\Delta V/\Delta t$)	3.6420	48.635
Voltage Unbalance (VU)	0.0045	0.9055
Total harmonic distortion (V_{THD})	0.4590	2.2435

The SFS method is initiated when system parameter values are between TH_L and TH_U (i.e. $TH_L < x_i < TH_U$). The OFP/UFP scheme is implemented in accordance with IEEE Std. 929-2000 which recommends any non-islanding inverter to cease to energize the utility lines within 6 cycles when the frequency is outside the range of 59.3-60.5 Hz [18].

Performance of the proposed method is evaluated based on simulation of system events under various levels of real and reactive power mismatch scenarios recommended as test

conditions in IEEE Std. 1547 [1], IEEE Std. 929 [18] and UL1741 [19]. The load and capacitor switching events were simulated by switching a 1MW/50kVAR load and a 70kVAR capacitor bank respectively. Islanding events were simulated by opening the circuit breaker CB to isolate the DG from utility grid. The accelerating gain parameter (k) and initial chopping fraction (cf_0) are set to 0.085 and zero respectively. Results from three test case scenarios are presented in this section.

A. Test Case Scenario I

Load L1 is adjusted to 100% of rated active power output of the DG unit ($P_{L1} = P_{DG}$) and zero reactive power ($Q_{L1} = 0$; unity power factor). Parameter values for islanding event at $t = 5s$, $x_1 = 86.75V$, $x_2 = 11.62V/cycle$, $x_3 = 0.812$ and $x_4 = 0.769\%$. Hence, SFS method is initiated as the parameter values are in the range $TH_L < x_i < TH_U$ and islanding is detected by OFP/UFP relay as system frequency deviates past the over frequency limit of 60.5 Hz. For load and capacitor switching events, the set of parameter values are $x_1 = 98.59V$, $x_2 = 18.17V/cycle$, $x_3 = 0.2412$, $x_4 = 1.305\%$ and $x_1 = 29.4V$, $x_2 = 5.107V/cycle$, $x_3 = 0.007$, $x_4 = 4.61\%$ respectively. In both cases, at least one of the parameter values is in the range $TH_L < x_i < TH_U$, hence SFS is activated. As the DG is grid-connected, injected perturbations do not cause significant deviations in system frequency.

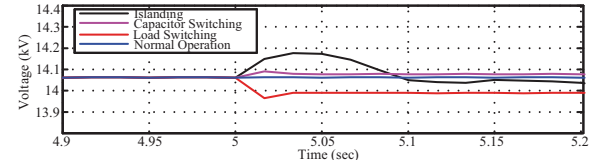


Figure 7. Voltage at PCC for system events in test case scenario-I.

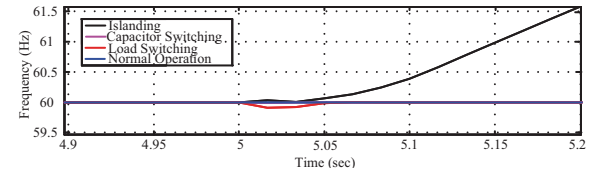


Figure 8. System frequency for system events in test case scenario-I.

B. Test Case Scenario II

Load L1 is set to 110% of rated active power of the DG unit ($P_{L1} = 1.1P_{DG}$) and reactive power is set to 1% of rated DG active power (Q_{L1} is positive; lagging power factor load). In case of islanding event at $t = 5s$, $x_1 = 191.02V$, $x_2 = 38.57V/cycle$, $x_3 = 0.271$ and $x_4 = 1.181\%$. Since, not all parameters are beyond TH_U , SFS is initiated and islanding is detected as the frequency deviates beyond over frequency set point. For load and capacitor switching events, $x_1 = 100.62V$, $x_2 = 18.25V/cycle$, $x_3 = 0.242$, $x_4 = 1.31\%$ and $x_1 = 27.24V$, $x_2 = 5.23V/cycle$, $x_3 = 0.007$, $x_4 = 4.58\%$ respectively. Hence, SFS method is initiated and perturbations are injected. As the DG is in grid-connected mode, system frequency does not deviate beyond permissible limits.

C. Test Case Scenario III

Load L1 is 125% of rated active power of the DG unit ($P_{L1} = 1.25P_{DG}$) and reactive power is set to 15% of the rated DG active power output (Q_{L1} is positive, lagging power

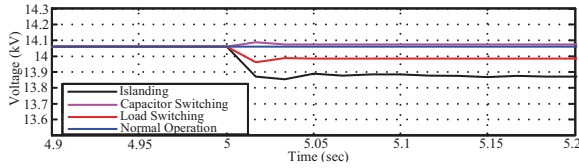


Figure 9. Voltage at PCC for system events in test case scenario-II.

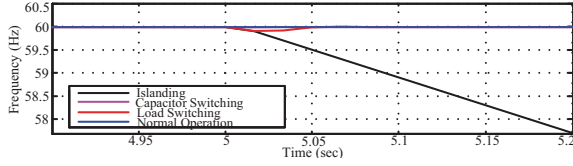


Figure 10. System frequency for system events in test case scenario-II.

factor load). For islanding event at $t = 5s$, $x_1 = 1965.26V$, $x_2 = 559.2V/cycle$, $x_3 = 5.97$ and $x_4 = 6.88\%$; which are all above threshold setting TH_U . Hence islanding is detected by the DT based passive scheme. For load and capacitor switching events, $x_1 = 112.78V$, $x_2 = 18.25V/cycle$, $x_3 = 0.241$, $x_4 = 1.31\%$ and $x_1 = 14.59V$, $x_2 = 5.27V/cycle$, $x_3 = 0.0068$, $x_4 = 4.54\%$. Hence, SFS method is initiated and perturbations are injected. As the DG is grid-connected, system frequency does not deviate beyond permissible limits.

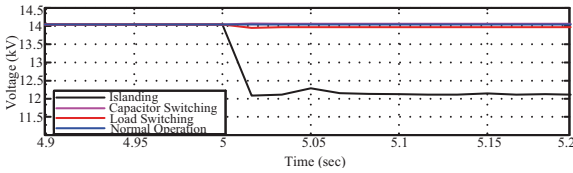


Figure 11. Voltage at PCC for system events in test case scenario-III.

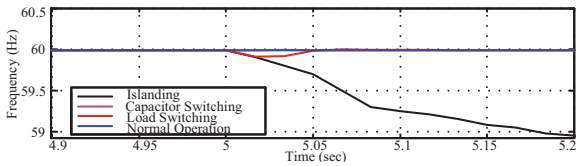


Figure 12. System frequency for system events in test case scenario-III.

Islanding detection speed for the proposed method is observed to be between 5 cycles or 0.08 seconds (detection by passive islanding relay) and 11 cycles or 0.18 seconds (detection by the combination of passive islanding relay and SFS method) which is within the detection time window specified by IEEE Std. 1547 [1].

V. CONCLUSION

A hybrid IDM for grid-connected inverter based DGs is presented in this paper. The proposed method integrates the advantages of DT based passive IDM and Sandia frequency shift method based active IDM. Case study results from a grid-connected PV system model indicate that the proposed hybrid scheme is capable of accurately detecting islanding events under various levels of real and reactive power mismatch between DG and local loads. Moreover, in the proposed method is capable of reducing power quality degradations by limiting

the perturbation injections by the SFS method. Furthermore, the proposed method relies completely on local measurements, thus eliminating the necessity of communication links. Further investigations may include performance assessment of the proposed method in presence of multiple inverter based DGs and optimization of active method to enhance overall performance of the proposed method.

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REFERENCES

- [1] IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems, IEEE Standard 1547.2-2008, 2009.
- [2] R. A. Walling and N. W. Miller, "Distributed generation islanding implications on power system dynamic performance," in *Proc. IEEE Power Eng. Soc. Summer Meeting*, 1985, vol.1, pp.92-96.
- [3] H. G. Far, A. J. Rodolakis, and G. Joos, "Synchronous distributed generation islanding protection using intelligent relays," *IEEE Trans. Smart Grid*, vol. 3, no.4, pp. 1695-1703, Dec.2012.
- [4] M.E. Ropp, M. Begovic, and A. Rohatgi, "Analysis and performance assessment of the active frequency drift method of islanding prevention," *IEEE Trans. Energy. Conv.*, vol. 14, no. 3, pp. 810-816, Sep. 1999.
- [5] G. A. Smith, P. A. Onions, and D. G. Infield, "Predicting islanding operation of grid connected PV inverters," in *Proc. IEE Electric Power Applications*, Jan. 2000, vol. 147, pp. 1-6.
- [6] W. Bower and M. Ropp, "Evaluation of islanding detection methods for utility-interactive inverters in photovoltaic systems," Sandia National Laboratories, New Mexico, Tech. Rep. SAND2002-3591, Nov. 2002.
- [7] H. H. Zeineldin and J. L. Kirtley, "Performance of OVP/UPV and OFP/UPF method with voltage and frequency dependent loads," *IEEE Trans. Power Del.*, vol. 24, no. 2, April 2009.
- [8] W. Freitas, W. Xu, and Z. Huang, "Comparative analysis between ROCOF and vector surge relays for distributed generation applications," *IEEE Trans. Power Del.*, vol. 20, no. 2, pp. 1315-1324, Apr. 2005.
- [9] S.K. Salman, D.J. King, and G. Weller, "New loss of mains detection algorithm for embedded generation using rate of change of voltage and power factors," in *Proc. Int. Conf. Devel. Pow. Sys. Prot.*, pp. 82-85.
- [10] S. Jang and K.H. Kim, "An islanding detection method for distributed generations using voltage unbalance and total harmonic distortion in current," *IEEE Trans. Pow. Del.*, vol. 19, no. 2, pp. 745-752, Apr. 2004.
- [11] K. El-Arroudi, G. Joos, I. Kamwa, and D. T. McGillis, "Intelligent based approach to islanding detection in distributed generation," *IEEE Trans. Power Delivery*, vol. 22, no. 2, Apr. 2007.
- [12] K. El-Arroudi, G. Joos, and I. Kamwa, "Data mining approach to threshold setting of islanding relays in distributed generation," *IEEE Trans. Power Systems*, vol.22, no. 3, Aug. 2007.
- [13] N. W. A. Lidula and A. D. Rajapakse, "A pattern recognition approach for detecting power islands using transient signals- Part I: design and implementation," *IEEE Trans. Pow. Del.*, vol. 25, no. 4, Oct. 2008.
- [14] V. Menon and M.H. Nehrir, "A hybrid islanding detection technique using voltage unbalance and frequency set point," *IEEE Trans. Power Sys.*, vol. 22, no. 1, pp. 442-448, Feb. 2007.
- [15] P. Mahat, Z. Chen, and B. B. Jensen, "A hybrid islanding detection technique using rate of change of voltage and real power shift," *IEEE Trans. Power Delivery*, vol. 24, no. 2, pp. 764-771, Apr. 2009.
- [16] H. Zeineldin and S. Kennedy, "Instability criterion to eliminate the non-detection zone of the Sandia frequency shift method," in *Proc. Power Systems Conf. and Expo.*, 2009, pp. 1-5.
- [17] I. Witten, E. Frank, *Data Mining: practical Machine Learning Tools and Techniques*. Morgan Kaufmann Publishers, 2011, pp. 365-483.
- [18] IEEE Recommended Practice for Utility Interface of Photovoltaic Systems, IEEE Standard 929-2000, April 2000.
- [19] *The standard for inverters, converters and controllers for use in independent power systems*, Underwriters Laboratories Standard 1741.